

## MS Access objects, data types and SQL ★ Golden

One college period · 40–45 minutes · Database fundamentals

### What you will learn

- List Access objects and data types for a long answer
- Write CREATE, INSERT, UPDATE, DELETE, SELECT
- Plan a Library Management schema

| Object         | Purpose                                |
|----------------|----------------------------------------|
| Table          | Store data in fields/records; set PK   |
| Query          | Select, update, calculate, join tables |
| Form           | On-screen data entry (like a VB form)  |
| Report         | Print marksheets, lists                |
| Macro / Module | Automate / VBA                         |

### MS Access data types (learn for Section C)

| Type                | Stores                              |
|---------------------|-------------------------------------|
| Text / Short Text   | Names, up to 255 chars              |
| Memo / Long Text    | Remarks, long notes                 |
| Number              | Byte, Integer, Long, Single, Double |
| Date/Time           | Dates and clocks                    |
| Currency            | Money, 4 decimal                    |
| AutoNumber          | Automatic PK                        |
| Yes/No              | Boolean                             |
| OLE Object          | Pictures, files                     |
| Hyperlink           | URLs                                |
| Attachment / Lookup | Newer Access extras                 |

```
CREATE TABLE Book (
  ISBN TEXT PRIMARY KEY,
  Title TEXT,
  Author TEXT,
  Price CURRENCY
);
INSERT INTO Book VALUES ('978-1', 'CS XI', 'STBB', 450);
SELECT Title, Author FROM Book WHERE Price > 400;
UPDATE Book SET Price = 500 WHERE ISBN = '978-1';
DELETE FROM Book WHERE ISBN = '978-1';
```

SELECT is projection (columns) + WHERE is selection (rows). JOIN combines tables on keys. Equijoin: WHERE Book.AuthorID = Author.ID. Operators: AND OR NOT, = <> > < LIKE BETWEEN.

### Planning a library database

1. Purpose: issue and return books.
2. Tables: STUDENT(Roll PK, Name, Class), BOOK(ISBN PK, Title, Author), ISSUE(IssueID PK, Roll FK, ISBN FK, IssueDate, ReturnDate).
3. Unique fields: Roll, ISBN, IssueID.
4. Relationships: Student 1—M Issue M—1 Book, all up to 3NF (no repeating books inside student).
5. Refine: add Fine, status Issued/Returned.

#### HOW BIEK ASKS THIS

Practical 2026 Access: create Library\_Management, PK, 5 records, query currently issued books.  
Theory: explain all Access data types.

#### اردو میں سمجھیں

Access میں Table ڈیٹا، Query سوال، Form انٹری، Report پرنٹ، SQL: SELECT چننا، INSERT داخل، UPDATE بدل، DELETE مٹانا۔

#### CHECK YOURSELF

1. Which object is for printing?
2. AutoNumber is typically used as?

Answers: 1. Report · 2. Primary key

### MCQs

1. MS Access object used to display and print data:
  - a. Report
  - b. Table
  - c. Form
  - d. Macro
2. This key uniquely identifies a record:
  - a. Super Key
  - b. Foreign Key
  - c. Conditional Key
  - d. Primary Key

Key: 1=a, 2=d

### Short questions (3 marks each)

1. Write SELECT syntax with WHERE.
2. Differentiate form and report.
3. Write steps to design a library database.

### Model short answers

1. SELECT field1, field2 FROM table WHERE condition; example SELECT Title FROM Book WHERE Price>400;
2. Form is interactive on screen; report is formatted for paper/PDF.
3. Purpose, tables, fields, unique keys, relationships to 3NF, refine.

### Long questions (board Section C)

1. Explain all the data types used in MS-Access. OR Explain SQL DML statements with examples.